Lake Geneva and the Geneva Lake shore, Wisconsin

El Niño reaches Lake Geneva as a warm, low-snow winter, and the danger is not the cold. It is a lake that freezes at the edges and stays open in the middle. In the last strong El Niño, cars went through the ice and a snowmobiler drowned in open water nobody had flagged. The most important step is to tell people, at the launches where they set off, whether the middle of the lake is open.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Walworth County, WI (inferred)	None published for the US, Wisconsin, or the county	n/a here; global 6-month figure 239,000 ± 11,000	Thin, partly frozen lake ice; then falls on glaze ice and outage-season CO

14/14 recommendations checked. 11 corrected, 1 confirmed as written (the “unreliable winter” framing), 1 unverified (parametric cover, which the panel already advised against), and 1 contraindicated (NFIP flood cover sold as an El Niño life-safety step). None of the eleven operational actions survived exactly as proposed. The four funding placements were checked separately: 1 contraindicated (a motel overflow for single adults through a shelter that serves families) and 1 unverified (a LIHEAP bridge float with no confirmed route in). **The funding target fell 73%**, from $301k to $80k, well past the one-third flag. Most of the cut came from furnace money that duplicates WHEAP's existing furnace programme.

* Suggested formulation: check with a health worker before sharing.

EXECUTIVE SUMMARY — AT A GLANCE



42.59°N 88.43°W, United States of America · Natural Earth

- **The risk.** The danger is a lake that freezes at the edges and stays open in the middle: in the last strong El Niño, cars went through the ice and a snowmobiler drowned in open water nobody had flagged.
- **First action.** GLLEA and Walworth County Emergency Management post a weekly open-centre ice status at every public launch, in English and Spanish, with boards up by 1 December.
- **The money.** $80k floor and target, about $226k ceiling; $30k buys rescue equipment on the lake against a written needs list, and $50k funds heating crisis aid.
- **Do not.** Do not sell flood insurance as an El Niño winter life-safety step.

What to do	**8** actions	p. 2
Who does what	RACI chart	p. 10
Where money does the most	funding	p. 10
Do not do this	**3** items	p. 12
What has happened here	the record	p. 13
How this was produced	methodology	p. 15
References	sources	p. 17

Exposure: Climate Impact Lab, 451,000 at Risk, September 2026, pre-peer-review. It publishes no figure for the United States, Wisconsin, or Walworth County, and heat is not this place's winter hazard. No population-scaled figure is given here, because scaling a heat number onto an ice hazard would be meaningless.

Read this first: the danger is a lake that half freezes, not a lake that stays open. All seven lenses reached this independently. Geneva Lake is 135 feet deep and freezes from the shallow ends inward, so the shore can hold while the middle is open water. That is what happened in February 2016, during the last strong El Niño: the warden described the lake as unfrozen "from Williams Bay almost to Fontana," and that is where Mark Reninger, 53, drove a snowmobile into open water at night. Eight days earlier, 15 cars parked on the ice at the public launch during Winterfest broke through. NWS Milwaukee's record backs the mechanism: all six strong El Niños were near or above normal temperature in southern Wisconsin, and five of six had near or below normal snow. One caveat cuts the other way. Lake drownings are rare and concentrated. The Public Health lens's point stands: the larger winter toll among older residents is probably falls on glaze ice, which death certificates record as falls, not weather.

WHAT THIS PLACE SHOULD DO RANKED

Actions are ranked by expected lives saved per unit of effort before the peak. The backing lenses are shown on each action, and the RACI chart below assigns each one. An action several lenses reached independently is the stronger bet, but only where its evidence came from somewhere like here. One rule underlies every lake action: the trigger is "is the centre open?", not "is there ice?"

1. Post a weekly open-centre ice status at every public launch, in English and Spanish, from the first skim ice

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · FRONTLINE · SKEPTIC | INVENTORY: WINTER WEATHER ADVISORIES (ANAND 2025), INDIRECT

OWNER	BY WHEN	COST	PREVENTS
Geneva Lake Law Enforcement Agency (GLLEA) with Walworth County Emergency Management; data from ice observer Theodore Peters and guide Doug Ide, with consent	Boards up by 1 Dec; updated weekly until ice-off	Under $3k for signs and QR boards (lens estimate, no quote)	Crossings over unflagged open water, as on 14 Feb 2016

* Suggested formulation: check with a health worker before sharing.

The data already exists and is not posted where people set off. Theodore Peters observes ice from shore at seven points, the basis of a record that begins in 1862 (Magnuson, UW, 2023). Doug Ide posts weekly auger readings near the Williams Bay launch. The board should state three things: whether the centre is open, the latest measured thickness at the launch, and that the Wisconsin DNR names no thickness as safe. Put it at the Lake Geneva, Williams Bay, and Fontana launches, in English and Spanish, and push the same text through County Emergency Management and DNR channels. The people on the ice in 2016 were visitors from Illinois, who do not see local warnings.

Transferability: the mechanism is local and documented, but no study shows that posted ice information reduces through-ice deaths, so treat this as unevaluated. Peters's observations record ice-on status, not safety. Up to five acres of open water still counts as “frozen” in his record. Ide is a private guide, so using his readings needs his agreement and a decision on who stands behind the board. That the 2016 open water was “unflagged” is the lenses' inference; it was not verified.

2. Close the public-launch ice to vehicles whenever the centre is open and on every event weekend, with signed parking on land

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · SKEPTIC · FRONTLINE · ECONOMIST · PUBLIC HEALTH | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
GLLEA board and the four shoreline municipalities; City of Lake Geneva council; VISIT Lake Geneva for event parking	**Ordinance or order by the December council cycle**	**Staff time and barricades; the Winterfest shuttle already exists**	**A repeat of 6 Feb 2016, and vehicles on ice over open water**

On 6 February 2016 about two dozen cars were parked on the ice at the public ice-fishing launch during Winterfest. Fifteen broke through, five were driven off, and ten were totalled. The only control was an “at your own risk” sign, and “hundreds park each year despite the signs.” Dispatch told callers “we can't do anything about the people on the lake.” **A Winterfest permit condition would not have stopped this**, because the parking was at the public launch, outside the event footprint, and VISIT Lake Geneva already ran a free shuttle. The lever is an icebound-waters rule: Wis. Stat. 30.81 lets municipalities regulate vehicles on icebound waters, and GLLEA's jurisdiction covers “the waters and ice of Geneva Lake.” No current Geneva Lake ordinance on vehicles on the ice was found.

* Suggested formulation: check with a health worker before sharing.

Transferability: same place, same failure. Winterfest itself is now shore-based (snow sculpting at Riviera Plaza and Flat Iron Park, 28 Jan – 1 Feb 2026), so the remaining exposure is the launch, not the event. No 2027 dates are published yet. Expect objection from ice anglers and tourism; the Skeptic's point is that closing the ice early, and being seen to do it, is the protective act.

3. Write the winter open-water rescue card now: name the hovercraft and airboats, their launch points, and dispatch on the first 911 call

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · PHILANTHROPIST · PUBLIC HEALTH | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Walworth County Emergency Management with MABAS Division 103; Lake Geneva FD, Town of Delavan FD, Williams Bay, Fontana, and Linn	Card agreed and tabletop run by 1 Dec	Within existing budgets; any equipment grant only against a written needs list	The hour-long wait that killed in 2016

On 14 February 2016 responders held back for about an hour because of open water mid-lake. The DNR warden said crews would not “put any emergency responders at risk” on it. Rescuers finally reached the men with a Town of Delavan hovercraft, and a Fox Lake, Illinois airboat was also sent. Reninger had been in the water 30–40 minutes. Cold-water physiology explains the stakes: incapacitation can come within 2–10 minutes in water near 0°C, and hypothermia follows over 30–60 minutes or more, so response time matters inside both windows. The fix is a written box card that names the open-water craft (Delavan's hovercraft and airboat, plus any other in reach), where each launches on the west end, and that they roll on the first call, not after a scene size-up. Stage them on event weekends and the first cold snaps.

* Suggested formulation: check with a health worker before sharing.

Transferability: same lake, same failure. Corrections from verification: Lake Geneva FD already runs a trained water and ice rescue team and trains every year in insulated suits, so the Philanthropist's $30k equipment grant may buy kit that already exists. Get a written gap list first. The January 2024 Linn rescue cited by a lens was a near-shore fall, with the woman pulled out by a Williams Bay officer, so it is not evidence for open-water assets. Whether a Division 103 water-rescue card already exists was not confirmed. Ask the division coordinator first.

4. Cut ice falls among people 65 and over: cleats for recent fallers, and Stepping On scheduled now

PARTIALLY CONFIRMED | PUBLIC HEALTH | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Walworth County ADRC	Cleats out by 1 Dec; Stepping On run in Oct–Nov	*Unfunded*, small; no quote for cleats	Outdoor falls on glaze ice, the larger hidden winter toll

Wisconsin has the highest fall-death rate of any state for people 65 and over: 158.4 per 100,000 in 2023, against 69.9 nationally (NCHS Data Brief 532). About 21.6% of Walworth County residents are 65 or older. In a Wisconsin trial, giving Yaktrax-type cleats to community-dwelling people 65+ who had fallen in the past year cut outdoor falls (RR 0.45; 0.42 on snow and ice days; McKiernan, *J Am Geriatr Soc* 2005, n=113). Exercise programmes cut the fall rate by about 23% (rate ratio 0.77, 59 trials, high certainty; Sherrington, Cochrane 2019). The ADRC already lists Stepping On, but its 2026 date is “TBA”. Schedule it for October and November, and recruit the same fallers who get cleats.

* Suggested formulation: check with a health worker before sharing.

Transferability: the cleat trial is from Wisconsin and this population, but it was one small, unblinded winter trial with self-reported falls, and it measured falls, not deaths. **No study links El Niño or warm winters to more ice falls**; the link is reasoning from freeze–thaw glaze, not evidence. Stepping On builds benefit over months, so classes started now protect mainly in late winter. The “more than 1,800 deaths in 2024” figure rests on secondary reporting and is not used here.

5. Get every disconnected or propane-heated household into WHEAP before 1 November

PARTIALLY CONFIRMED | ECONOMIST · FRONTLINE · PHILANTHROPIST | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Energy Services of Walworth County (Elkhorn); state line 1-866-432-8947; utilities for payment arrangements	**31 Oct for reconnections; propane households pre-enrolled by 1 Dec**	**Outreach time; benefits already appropriated (crisis max $1,200)**	**Households locked out of heat until 15 April**

The winter moratorium (PSC 113.0304) bars heating disconnections for nonpayment from 1 November to 15 April. But it only protects households that are connected when it starts, and utilities need not reconnect without a payment arrangement. So a household dark on 31 October can stay dark all winter. Propane households have no moratorium at all, because propane dealers are not regulated utilities. WHEAP's FY2026 crisis maximum is $1,200 and its regular heating maximum $2,147; Wisconsin's FY2026 LIHEAP allocation was $101.5M. The action is moving households across the 1 November date in the next three weeks.

* Suggested formulation: check with a health worker before sharing.

Transferability: same state and programme. A warm El Niño winter lowers the stakes but does not remove them; NWS notes days below normal still occur. Corrections: the local agency is Energy Services of Walworth County, not county DHHS; its phone number came from an old page and should be checked. Walworth's propane share of heating was not obtained (census API unavailable to the verifier); the statewide figure is roughly 12% (secondary source). FY2027 LIHEAP is running on a continuing resolution to 11 December 2026.

6. Check, test, and replace CO alarms in older and rental housing, and say plainly: never run a generator in the garage

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Lake Geneva FD and Williams Bay and Fontana FDs, with the buddy-list volunteers (action 7)	**Before 1 Dec, ahead of ice-storm outages**	***Unfunded*, small; no unit cost found**	**Carbon monoxide poisoning during outages**

In Wisconsin Poison Center data for 2014–2016 (n=408), people with no detector had 3.2 times the odds of CO poisoning compared with people whose detector alarmed (95% CI 1.5–6.9). Wisconsin already requires CO alarms in homes (Wis. Stat. 101.647 and 101.149), so this is not a distribution campaign. It is a check for dead, expired, or missing alarms, aimed at older and rental housing, paired with the generator message before the first ice-storm outage. Wisconsin's CO emergency-visit rate runs 5.7–11.2 per 100,000.*

* Suggested formulation: check with a health worker before sharing.

Transferability: Wisconsin data. State alarm laws show no measurable effect on CO deaths (Hampson 2016; *Clin Toxicol* 2023), which is why the action is a door-to-door check rather than more law. The lens's "about 450 ER visits a year" could not be verified and is replaced with the published rate.

7. Run a volunteer cold-snap and outage buddy list from the food pantry and St. Francis de Sales

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · SKEPTIC | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Lake Geneva Food Pantry and St. Francis de Sales parish volunteers; Walworth ADRC for referrals	**List built by 1 Dec**	**Volunteer time; *unfunded*, small**	**Isolated older residents and power-dependent residents going unchecked in an outage**

The Lake Geneva Food Pantry at 715 Wisconsin St has been volunteer-run since 1983 and serves about 414 people a month, rising past 580 in December. St. Francis de Sales has a Hispanic Prayer Group and Spanish Mass. The Archdiocese lists 1st and 3rd Mondays at 6pm, while one directory says Sundays at 5pm, so call (262) 248-8524 to confirm. Volunteers sign up older people living alone along the shore roads, among the empty seasonal homes, and anyone on oxygen or powered medical equipment, then phone or visit them during cold snaps and outages. The pantry table can also send households to WHEAP from October, since the season opened 1 October. HHS emPOWER gives monthly counts of Medicare beneficiaries on powered equipment by ZIP, so ZIP 53147 can be pulled to size the list.

* Suggested formulation: check with a health worker before sharing.

Transferability: the isolation evidence is from heat, not cold (Semenza, *NEJM* 1996, Chicago: living alone adjusted OR 2.3). In Wisconsin's 2014 hypothermia deaths, some indoor victims had furnaces off or broken. No trial of check-ins exists. Low cost, plausible, unproven. Latino share of the City of Lake Geneva is 15.6% (2020 Census), not the 17% the lens took from 2010. No Latino-specific nonprofit was found in the county, which is not the same as none existing; check with United Way 2-1-1.

8. Issue an early-December thin-ice advisory for Lake Como and Delavan Lake after any warm fall

PARTIALLY CONFIRMED | HISTORIAN | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Walworth County Sheriff and Emergency Management; Town of Geneva for Lake Como	**First week of December**	**Staff time**	**Early-season through-ice deaths on the shallow lakes, which freeze first**

Michael Montemayor, 44, of Burlington walked out onto Lake Como from Geneva National and went through. His body was recovered on 9 December 2016, and Sheriff Picknell cited an “unusually warm fall and early winter.” The Town of Geneva already has an icebound-waters ordinance for Lake Como, which is the natural place to hang an advisory.

Transferability: correction from verification. Winter 2016–17 was a weak La Niña, not an El Niño, so this case shows the warm-fall mechanism rather than an El Niño precedent. Delavan Lake was not checked.

How to say it: an “unreliable winter,” not a “mild El Niño winter.” Confirmed as written. Say: likely milder and less snowy on average, but expect dangerous cold snaps and unsafe ice. Do not claim El Niño causes the cold snaps. NWS Milwaukee notes the shifted jet stream “limits cold air intrusions” without ending them, and January 2024's Arctic outbreak arrived during a strong El Niño.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Geneva Lake Law Enforcement Agency	City of Lake Geneva	Williams Bay, Fontana & Linn	Walworth County Emergency Mgmt	Lake Geneva Fire Dept	Town of Delavan FD & MABAS Div 103	Wisconsin DNR wardens	VISIT Lake Geneva	Ice observers (Peters, Ide)	Walworth County ADRC	Energy Services of Walworth Co.	Utilities	Food Pantry & St. Francis de Sales	Walworth Co. Sheriff & Town of Geneva
1. Weekly open-centre ice status at launches	**A**	**R**	**R**	**R**	**C**		**C**	**I**	**R**				**I**	**I**
2. Close launch ice to vehicles when centre open	**A/R**	**R**	**R**	**C**	**C**		**C**	**C**	**C**					**I**
3. Winter open-water rescue card	**C**	**I**	**R**	**A**	**R**	**R**	**C**	**I**						**R**
4. Cleats and Stepping On for 65+ fallers					**C**					**A/R**			**C**	
5. WHEAP before 1 Nov; propane households		**I**								**C**	**A/R**	**R**	**R**	
6. CO alarm check and generator message		**C**	**R**	**C**	**A/R**					**C**		**I**	**R**	
7. Cold-snap and outage buddy list		**I**		**C**	**I**					**C**	**C**	**C**	**A/R**	
8. Small-lake thin-ice advisory	**I**			**R**	**I**	**I**	**C**	**I**	**C**					**A/R**

What the chart shows. **GLLEA is accountable for both ice-access rows**, because its jurisdiction covers the ice of the whole lake and no single town owns it; that is also why the shoreline villages carry an R. **The rescue row needs the county, not the city**: the craft that reached the victims in 2016 came from the Town of Delavan and Fox Lake, Illinois, outside every shoreline municipality. **Row 7 has no public owner**: the buddy list is volunteer work, so its continuity depends on the pantry and the parish. **The ice observers carry an R on row 1 without any agreement to do so**; their consent is the row's first dependency.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. Most protective actions here cost staff time, not money.

FLOOR

$80k

TARGET

$80k

CEILING

~$226k

LARGEST UNCERTAINTY

No local count of cold or ice deaths exists, so the baseline may be close to zero

* Suggested formulation: check with a health worker before sharing.

Floor and target: $30k for rescue equipment on the lake, released only against a written needs list, plus $50k for heating crisis aid after WHEAP crisis money runs out, which totals $80k. **Ceiling:** $80k, plus $96k for winter shelter for single adults if a provider that actually serves them is confirmed, plus $50k for a LIHEAP bridge float if a route into WHEAP is confirmed, which totals $226k. The target fell 73% from the lens's $301k. The $75k furnace line duplicates WHEAP's furnace programme and the Keep Wisconsin Warm/Cool Fund, and the $96k shelter line was aimed at a family shelter.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Lake Geneva, Williams Bay, and Fontana fire departments	Action 3: cold-water suits, rescue sleds, throw bags; thin-ice signs at ~20 access points	$30k	Direct grant, conditional on a gap list	Equipment costs assumed; no published cost per death. LGFD already owns and trains in insulated suits	PARTIALLY CONFIRMED
Energy Services of Walworth County or Keep Wisconsin Warm/Cool Fund	Action 5: crisis heating aid for no-heat households after WHEAP crisis money runs out	$50k (was $125k)	Top-up to existing scheme	100 households × $500. Chirakijja, Jayachandran & Ong: cheaper heating cut US winter mortality 1.6% in gas-heated homes. The lens's “$1.9M per death” was its own division and is not used	PARTIALLY CONFIRMED
To be confirmed	LIHEAP bridge float if federal funds lapse after the 11 Dec continuing resolution	$0, held ($50k contingency)	Recoverable float	No route found for private money into WHEAP; ask DOA in writing first	UNVERIFIED
~~Community Action Inc. of Rock & Walworth (Twin Oaks)~~	~~Motel overflow for single adults, 10 × 120 nights × $80~~	~~$96k~~	n/a	Twin Oaks is “Walworth County's only 24/7 homeless shelter for *families*”; no motel programme found; $80 rate assumed; no El Niño link	CONTRAINDICATED

The funder's one thing: generic warming-centre money is wasted in a warm winter here. The cheapest life saved is on mid-winter open water after the summer-only Water Safety Patrol has gone home. Even that is mostly a written rescue card and a sign, not a grant. Most of what saves

* Suggested formulation: check with a health worker before sharing.

lives on this lake is free.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for this hazard and would cost lives or waste the window here.

Sell flood insurance as an El Niño winter life-safety step

CONTRAINDICATED AS FRAMED

The Economist proposed buying NFIP cover by about 7 November, ahead of the 30-day wait, against rain-on-snow flooding. But Walworth's documented floods are summer convective events: July 2017 hit 175 homes and did about $2.6M of infrastructure damage, and the historical record turned up no winter flood. A warm, low-snow El Niño winter lowers snowmelt and rain-on-snow risk. And insurance pays for property, not lives.

Instead: keep flood cover as an ordinary summer-resilience decision, and spend the autumn on the 1 November WHEAP deadline (action 5), which is the date that actually locks people in.

Fund motel overflow for single adults through Twin Oaks

CONTRAINDICATED AS SPECIFIED

The Philanthropist's $96k placement assumed Twin Oaks in Darien is the county's only fixed shelter and serves single adults. Community Action describes it as the county's only 24/7 shelter *for families*. A rotating church men's shelter existed in 2016, and no current single-adult provider or motel programme was found. The need may be real, but it is year-round, not an El Niño effect, and a warm winter lowers the exposure.

* Suggested formulation: check with a health worker before sharing.

Instead: find out who shelters single adults in the county this winter, through United Way 2-1-1, before placing any money.

Publish an "X inches is safe" rule *

AVOID (SKEPTIC; SUPPORTED BY DNR POSITION, NOT SEPARATELY VERIFIED)

The Wisconsin DNR gives no hard thickness standard. Its off-highway-vehicle lead says "we really shy away from saying that there's ever any ice that's 100% safe," and called last winter "one of the more dangerous years on ice that I can remember." Freeze–thaw produces layered ice that a single auger hole cannot read, and a measured six inches at the launch says nothing about the open centre.

Instead: publish the measured reading *and* whether the centre is open (action 1), alongside the DNR's position that no ice is safe.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1862	Geneva Lake ice record begins	n/a	One of the longest lake-ice records in the US
"1997"	First year with no recorded ice-on; the source does not say which winter, possibly the 1997–98 strong El Niño	None recorded	Treated as a curiosity
2001, 2012, 2017, 2020	Lake did not fully freeze	None recorded	Nothing
6 Feb 2016	Strong El Niño: ~24 cars parked on the ice at the public launch during Winterfest; 15 broke through	No injuries; 10 cars totalled	Winterfest later moved to shore; launch parking unregulated
14 Feb 2016	Snowmobile into open water between Williams Bay and Fontana, about 9pm	1 dead (Mark Reninger, 53); 1 hypothermic survivor	No change to rescue staging found

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
9 Dec 2016	Lake Como, after an “unusually warm fall” (weak La Niña)	1 dead (Michael Montemayor, 44)	Town of Geneva ordinance already in place
July 2017	Summer flood	175 homes; ~$2.6M infrastructure damage	Outside this winter window
Jan 2024	Strong 2023–24 El Niño: lake still open in mid-January (lens report); 28 Jan near-shore fall through ice in Linn	Woman rescued by a Williams Bay officer	n/a
Jan 2026	Winterfest held on shore (Riviera Plaza, Flat Iron Park)	n/a	Event exposure removed; launch exposure remains

The recurrence nobody has noticed: the lake now fails to freeze fully about one winter in five, and every one of those winters has been reported as a curiosity (“the lake didn't freeze”). Nobody has turned a 160-year ice record into a public warning. The winters that kill are not the ones where the lake stays open; they are the ones where it half freezes.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **The lake is the threat** (Emergency manager, Historian, Skeptic, Frontline, Philanthropist) versus **ice falls among older people are the threat** (Public health). The lake deaths are few, documented, and concentrated in a known group. Falls are many, diffuse, and invisible as weather deaths. Both are kept, ranked 1–3 and 4. Settling it needs a Walworth fall-death count by month from WISH, which no lens could pull.
- **A warm winter lowers cold risk** (Public health, NWS) versus **“mild” framing breeds complacency** (Skeptic). Verification sided with both: say “unreliable,” not “mild,” and do not claim El Niño causes the cold snaps.
- **Fund heating aid** (Philanthropist) versus **a warm El Niño winter weakens the case and WHEAP already pays** (finance verifier). The placement survived at $50k, aimed only at households past the WHEAP crisis cap.
- **Close the ice** (Skeptic, Emergency manager) versus **the tourism and ice-fishing economy**, which the Skeptic expects to object. No lens costed the economic loss of a closed launch.

* Suggested formulation: check with a health worker before sharing.

WHAT NOBODY COVERED — BLIND SPOT

No lens looked at **road crashes on glaze ice**, which are likely the largest winter killer in a county crossed by I-43 and US-12 and which freeze–thaw winters make worse. No lens looked at **ice-shanty heaters** as a CO source on the lake itself. No lens pulled the **emPOWER count** for ZIP 53147, so the number of power-dependent residents is unknown. No lens covered **seasonal homeowners**: burst pipes in empty houses are a property issue, but the elderly year-round residents living among them are the isolation risk the buddy list is meant to reach.

HOW TO READ THIS — METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people, and are not neighbourhood figures. Walworth County holds about 106,000. The El Niño projection covers *heat only*; ice, flood, cold, and outages are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The location was taken as given: the City of Lake Geneva and the Geneva Lake shoreline (Williams Bay, Fontana, Town of Linn), Walworth County, Wisconsin. Climate Impact Lab impact region, as this report records it: Walworth County, WI, *inferred* rather than resolved on the Impact Map. Mortality figures exist only at that scale; lake-level detail is used for hazard exposure and action design. Where lenses used administrative data, the unit is named: Wisconsin (fall and cold deaths, CO rates), Walworth County (age structure), and the City of Lake Geneva (2020 Census ethnicity). None is the shoreline itself, and visitors from Illinois, who made up both 2016 incidents, appear in none of them.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Heat is not this place's hazard, so the historical record carries the hazard instead of a borrowed number.

* Suggested formulation: check with a health worker before sharing.

- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias before it counted. Here the seven-lens convergence on the ice rests on local incident records, not a shared famous study, so it held; the six-lens “event permit” version was corrected because the 2016 parking was outside the event. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six separate verifier agents checked 14 recommendations and the four funding placements on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not re-opened in this run: a verifier loaded the page but its entries render only in a browser. The inventory's US filter (19 entries) was read in a browser earlier the same day for the Gaslamp Quarter briefing, and those titles were used here. Only *Winter weather advisories* (Anand 2025) bears on any action, indirectly on action 1 and the messaging note; it concerns vehicle crashes after snow advisories, not ice. No US entry covers lake-ice access, ice rescue, fall prevention, or CO alarms, so those actions are tagged “none found”.
- **Health wording.** Passages where a wording error could hurt someone (carbon monoxide, and lake ice) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total without the division being shown and labelled. The lens's $1.9M-per-death heating figure was such a division and is excluded. No local figure is invented; where data is thin, the report says so.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The shared web-search budget ran out during verification, so some checks were made by fetching pages directly, and genevalakeagency.org did not resolve. No Walworth count of cold, fall, or ice deaths was found. Walworth's heating-fuel mix was not obtained. Whether a MABAS Division 103 water-rescue card exists was not confirmed. The 2016 incident figures rest on TV-station reporting rather than agency records.

* Suggested formulation: check with a health worker before sharing.

References · verified by Phase 5 unless marked. 2016 snowmobile death, CBS Chicago (verified) · 2016 rescue delay, NBC26 (verified) · Delavan hovercraft, HNGN (verified) · Winterfest 911 calls, FOX6 (verified) · Winterfest cars, CBS58 (verified) · Winterfest shuttle, ABC7 (verified) · Winterfest 2026 (verified) · GLLEA jurisdiction (verified) · Magnuson, Geneva Lake ice record (verified) · Ice measurements, TMJ4 (verified) · Lake Como 2016, CBS58 (corrected) · Lake Como icebound-waters ordinance (verified) · Linn rescue 2024, FOX6 (corrected) · Lake Geneva FD (verified) · LGFD ice training, CBS58 (verified) · Cold-water 1-10-1 (verified) · DNR on ice safety, Wisconsin Watch (lens, not re-verified) · NWS Milwaukee, El Niño and southern Wisconsin (verified) · McKiernan 2005, cleats (corrected) · Sherrington 2019, Cochrane (verified) · NCHS Data Brief 532, fall deaths (corrected) · Walworth ADRC prevention programmes (verified) · CO detector effectiveness, Wisconsin (verified) · Wis. Stat. 101.149 (verified) · Wisconsin DHS CO data (corrected) · PSC winter moratorium (verified) · LIHEAP FY2026 release (verified) · FY2027 continuing resolution, CRS (verified) · WHEAP (verified) · WHEAP furnace component (verified) · Keep Wisconsin Warm/Cool Fund (verified) · Chirakijja, Jayachandran & Ong (verified) · Twin Oaks shelter (corrected) · July 2017 flood, CBS58 (verified) · Semenza 1996 (verified) · HHS emPOWER map (verified) · St. Francis de Sales (verified) · Lake Geneva Food Pantry (verified) · City of Lake Geneva 2020 profile, SEWRPC (verified) · Wisconsin cold deaths (verified)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (US)
Generated 8 October 2026 for Lake Geneva and the Geneva Lake shore, Wisconsin. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.